PROJECT-BASED LEARNING

PHASE-I EVALUATION

Project Title

TeamForge - Skill-Based Hackathon & Project Team Matchmaker

Team ID: DSCPP-III-2026-T238

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun

Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T238
Semester: III
Section: ML2, ML1, DS2, ML3
Domain: DSA + OOP with C++

Team Members

1. Mukul Fartiyal – 2510370466
2. Anubhav Bisht – 2510370178
3. Siddhant Singh – 2510380233
4. Mehul – 2510370093

Mentor

Dr. Hriday Kumar Gupta **Interactions completed:** 01

Problem Statement & Motivation

Problem Statement

Students often form teams for hackathons, PBLs and projects through friends or informal groups. This can create repeated skills, missing roles and uncertainty about contributions. TeamForge aims to help students find complementary teammates through skills.

Motivation

- Team formation is often informal and friend-based.
- Students may not know each other's technical strengths.
- Important skills or roles may be missing.

Target Users / Stakeholders

- Students looking for teammates for projects or hackathons.
- Project / hackathon organisers needing skill coverage.
- Teams that want to identify skill gaps before starting work.

Objectives

Project Objectives

1. Create student profiles containing skills offered, skills wanted, experience and availability.
2. Allow a project or hackathon to define required skills.
3. Retrieve and rank suitable candidates using suitable data-structure based methods.
4. Use C++ OOP to keep the application modular and extendable.
5. Show team coverage, remaining skill gaps and match scores.

Key Objective: Help students form balanced teams by matching complementary skills instead of only matching similar profiles.

Proposed Solution

Proposed Approach

1. Student creates a skill profile.
2. Project owner enters required skills.
3. System retrieves relevant candidates.
4. Candidates are scored and ranked.
5. Team options are checked for skill gaps.
6. Selected team details can be saved locally.

Key Features

- Student profile and requirement management.
- Skill-based search and top-K ranking.
- Team formation and coverage view.
- Skill-gap identification.
- Team history / local persistence.

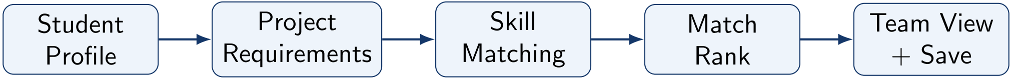
The first version focuses on a simple, explainable matching workflow that can be developed and tested locally.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	Searching, hashing, queues / priority queues, trees and graphs	Matching, retrieval and ranking
OOPs with C++	Classes and objects, constructors, inheritance, polymorphism, exceptions and file handling	Application structure and local storage

Requirement: Demonstrate meaningful integration of both subjects in one project.

System Architecture / Workflow



Major Components

- Profile Manager + Requirement Manager
- Matching Engine + Ranking
- Team Manager + local storage

Data / Control Flow

- Profile →skills →candidate retrieval
- Candidate set →score →ranking
- Team selection →coverage check →save

Technology Stack & Methodology

Technology Stack

- Programming: C++
- UI Framework: Qt
- Storage: Local structured files
- IDE: VS Code / Qt Creator
- Version Control: Git / GitHub

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

Qt was selected for the desktop UI after mentor discussion. It provides the interface tools needed for the C++ application.

Initial Progress & Team Contribution

Progress Achieved

- Discussed multiple project ideas with the mentor.
- TeamForge was approved on 31 Aug 2026; problem, users and features were fixed.
- User flow and architecture were drafted.
- C++ with Qt was selected; initial matching was discussed.

Individual Contribution

Member	Role	Contribution
Mukul Fartiyal	Team / Integration Lead	Requirements, architecture and co-ordination.
Anubhav Bisht	UI / Testing / Documentation Lead	Qt UI planning, testing plan and documentation.
Siddhant Singh	C++ / OOP Lead	C++ class design and application structure.
Mehul	DSA Lead	Matching, searching, hashing and ranking.

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Topic approval, requirements and design.
2. Phase-II: Application and matching development.
3. Phase-III: Testing, refinement and final demonstration.

Expected Outcomes

- C++ / Qt desktop prototype.
- Ranked teammate recommendations.
- Coverage + local storage.

References

1. Qt Documentation, Qt Widgets and Layouts.
2. cppreference.com, C++ Standard Library Reference.
3. Reema Thareja, *Data Structures Using C*, Oxford University Press.
4. E. Balagurusamy, *Object Oriented Programming with C++*.

Thank You
Questions & Suggestions